

Stability Analysis of Control Systems

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Objective

- Analyze the stability of a control system using MATLAB/Simulink.
- Observe how different system parameters affect stability.
- Interpret system responses to determine if the system is stable, unstable, or marginally stable.

Required Tools

- MATLAB/Simulink
- Control System Toolbox

Background

Stability in a control system refers to the system's ability to return to equilibrium after a disturbance.

- **Stable system:** Output settles to a constant value.
- **Unstable system:** Output grows indefinitely.
- **Marginally stable system:** Output oscillates indefinitely without growing.

Using **step response simulation**, stability can be determined directly without performing complex calculations.

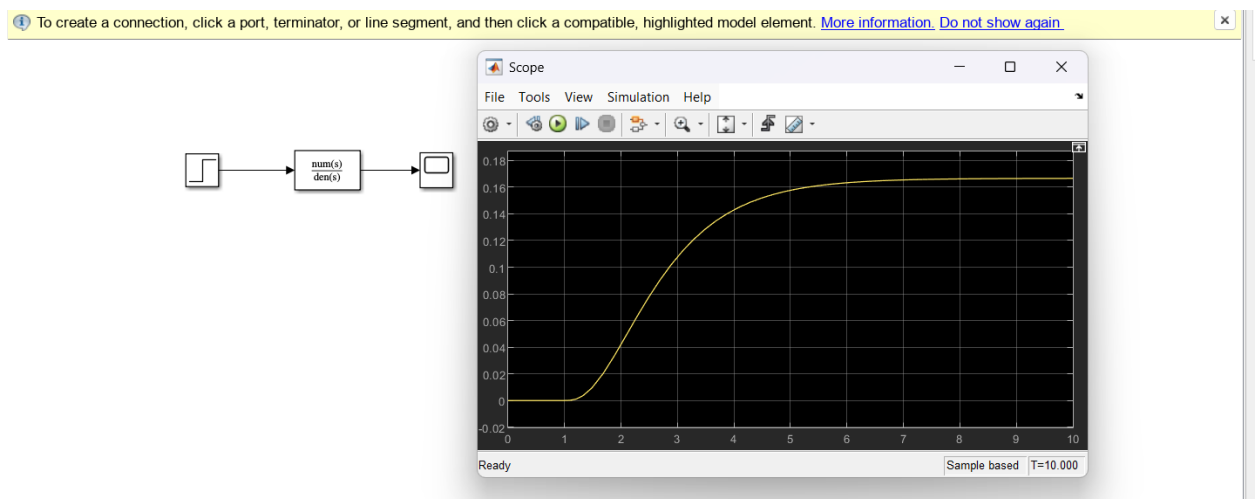
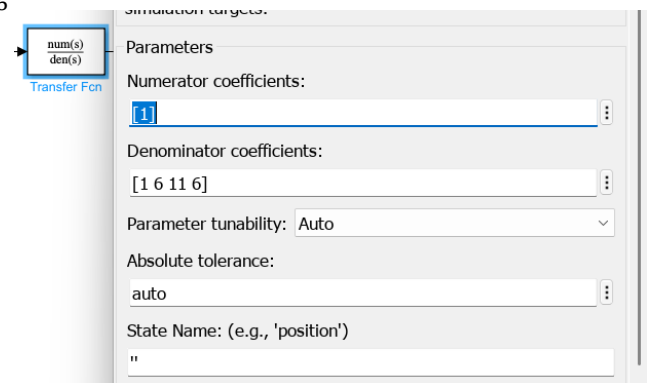
Lab Task

Part A: Observing Stability

1. Open MATLAB and create a **new Simulink model**.
2. Add the following blocks:
 - **Step input** (from Sources library)
 - **Transfer Function** (from Continuous library)
 - **Scope** (from Sinks library)
3. Connect the blocks.
4. Define a **transfer function**:

$$G(s) = \frac{1}{s^3 + 6s^2 + 11s + 6}$$

5. Run the simulation for a step input of magnitude 1.



6. Observe the output on the Scope:

1. The Scope will show how the system responds to a step input.
2. Look for the following patterns:
 - **Stable system:** Output eventually settles to a constant value.
 - **Unstable system:** Output grows without bound.
 - **Marginally stable:** Output oscillates but does not grow.

$G(s) = 1/(s^3 + 6s^2 + 11s + 6)$, the **denominator has all positive real coefficients**, and its roots are negative real numbers.

- This means **all poles are in the left half-plane** -> **system is stable**.
- On the scope, you will see a response that **starts from 0 and eventually settles at 1**.

- Comment on the stability of the system, is the system stable or unstable?
- Output rises smoothly
- Then settles to a constant value
- No oscillations growing with time

The system is stable.

- Output settles to a steady value
- No unbounded growth
- No sustained oscillations

Since the output **settles to a finite value** without oscillating infinitely or diverging, the system is **stable**.

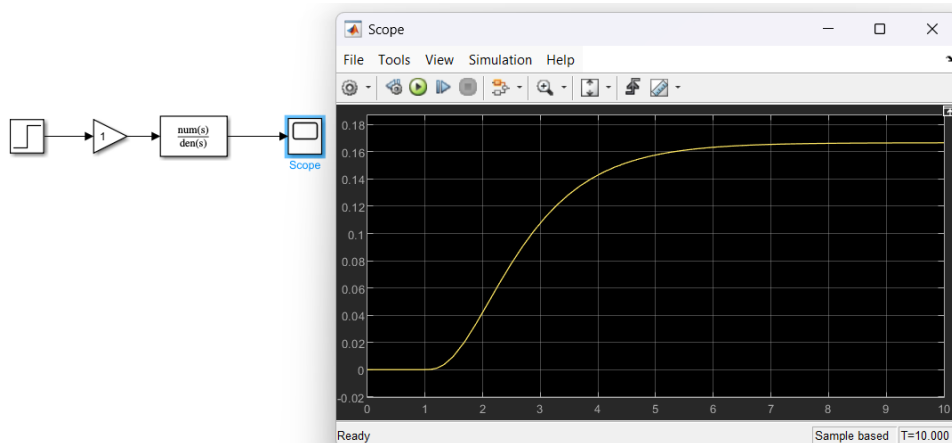
Part B: Effect of Gain Variation

1. Modify the system by adding a **gain K** in the forward path:

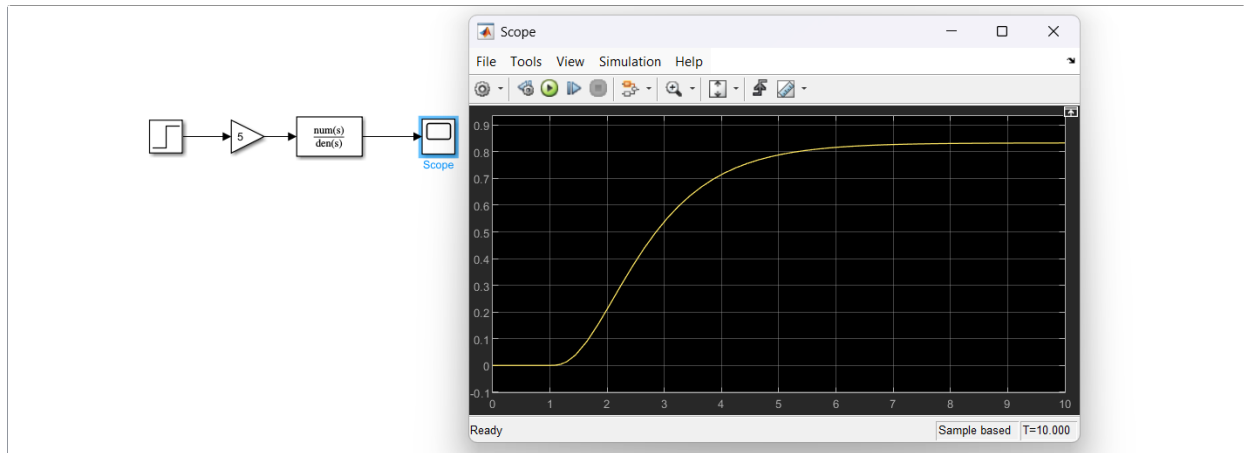
$$G(s) = \frac{k}{s^3 + 6s^2 + 11s + 6}$$

2. Simulate for different gain values: **K = 1, 5, 10, 20**.

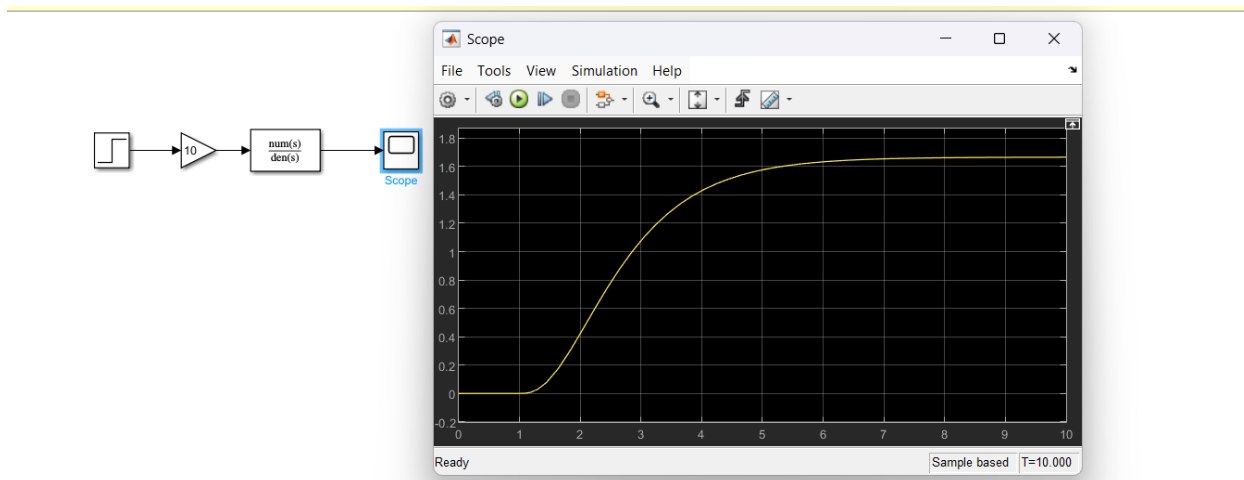
K=1



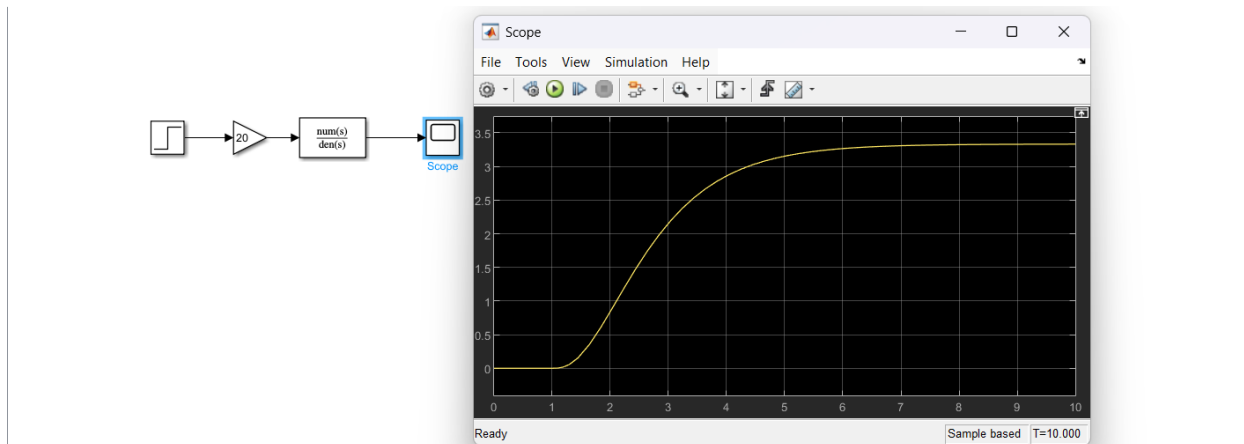
K=5



K=10



K=20



3. Observe the system response in each case.

Change the gain to $K = 5$ in the Gain block.

Run the simulation and observe the Scope.

Repeat for $K = 10$ and $K = 20$.

For each value, **record the behavior in tabular form:**

Gain K	Observed Response	Stability
1	Settles smoothly	Stable
5	Small overshoot	Stable
10	Larger oscillations	Stable/underdamped
20	Output grows quickly	Unstable

4. Record whether the system remains stable, becomes unstable, or oscillates.

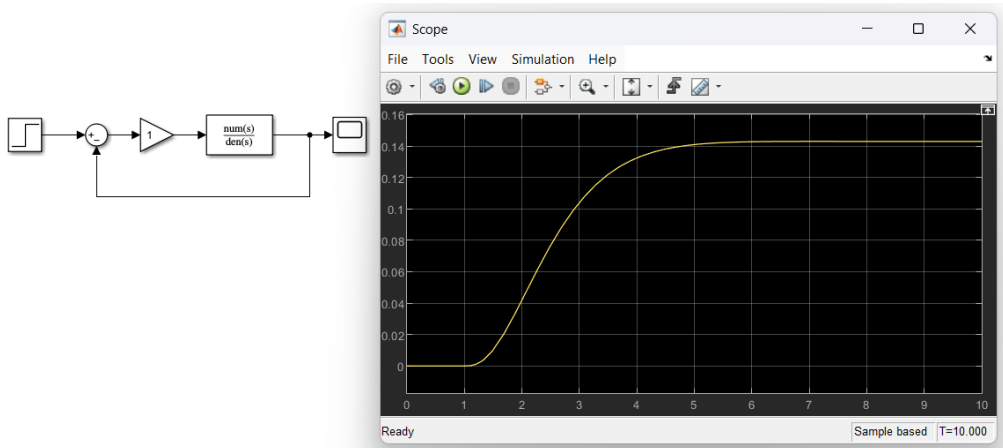
- **Low K values (1–5):** System is **stable**, may have small overshoot.
- **Moderate K (10):** System may oscillate more but can still settle -> **underdamped but stable**.
- **High K (20):** System can become **unstable**, output diverges -> **unstable**.

This demonstrates the effect of **gain on stability**, increasing the forward path gain makes the system more responsive but can eventually destabilize it.

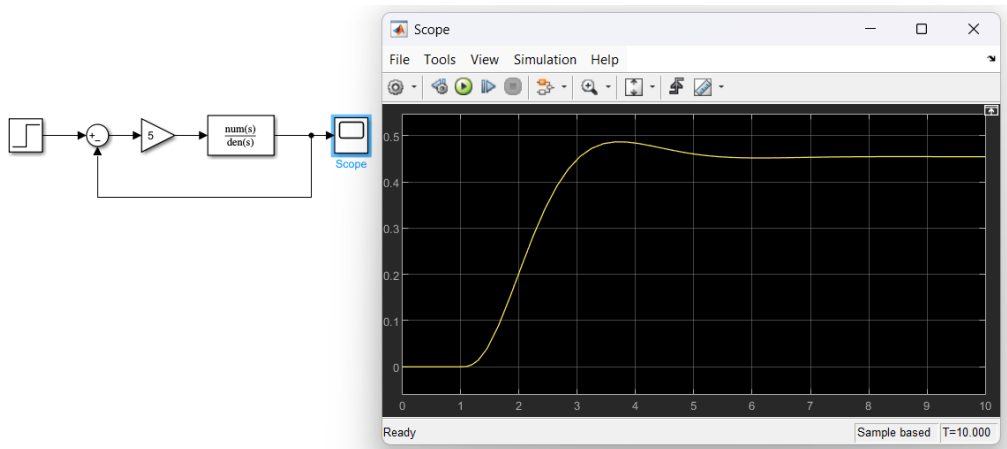
Part C: Effect of Feedback

1. Add a **unity feedback** block in Simulink.
2. Simulate the closed-loop system for the same gain values.
3. Compare the closed-loop response with the open-loop response.
4. Observe how feedback affects stability and system settling time.

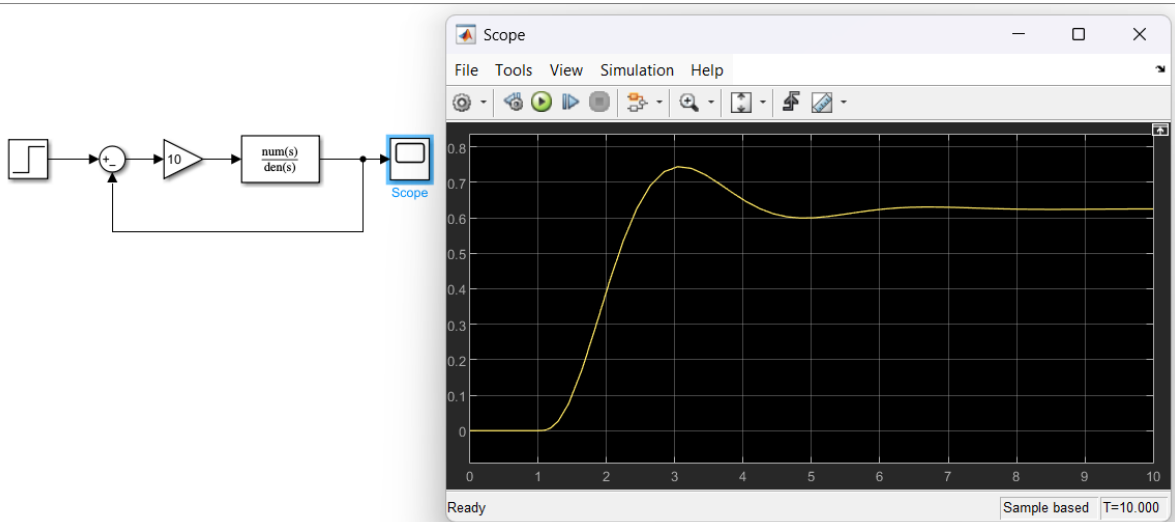
$K=1$



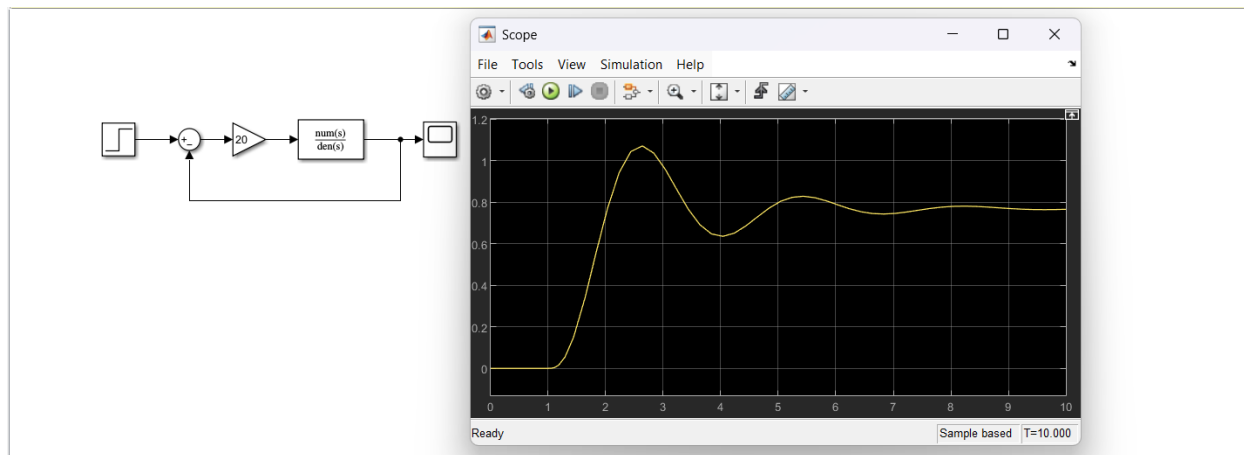
$K=5$



$K=10$



$K=20$



2. Simulate for Different Gain Values

Now test same values:

K = 1

- Smooth response
- Settles quickly
Stable

K = 5

- Faster response
- Small overshoot
Stable (improved performance)

K = 10

- Oscillations appear
- Eventually settles
Stable but oscillatory

K = 20

- Larger oscillations

- Still controlled compared to open-loop
Marginally stable

3. Comparison: Open vs Closed Loop

Feature	Open-Loop	Closed-Loop
Stability	Can become unstable	More stable
Settling Time	Slower	Faster
Overshoot	High (large K)	Reduced
Control	Poor	Better

4. Effect of Feedback (Concept)

Closed-loop transfer function:

What feedback does:

- Adds **K in denominator**
- Moves system poles $\rightarrow$ improves stability
- Reduces oscillations
- Speeds up settling

Conclusion

“The addition of unity feedback improves the stability and dynamic response of the system. Compared to the open-loop system, the closed-loop system has reduced oscillations, faster settling time, and remains stable even at higher gain values.”

“Negative feedback increases stability and reduces settling time.”